Department of Computer Science, University College London

COMP0222 / COMP0249 Lab 06: COLMAP

9th February, 2026

1 Scope and Purpose

This lab gives you some hands-on experience of using COLMAP and consists of three parts: installing COLMAP on your machine, testing it with some reference datasets, and then encouraging you to go, collect your own data, and see how it works. This lab does not directly contribute to Coursework 01. However, using COLMAP will form part of the knowledge you will need for Coursework 02.

2 Activities

Activity 0: Install COLMAP

Go to the COLMAP project page and install COLMAP. COLMAP will run natively on Windows, Mac and Linux.

The software installation page is at `https://colmap.github.io/install.html`. However, it is package managed as well. If you use a Linux machine, you can install it from:

```
sudo apt install colmap
```

If you use a Mac and use homebrew, you can use:

```
brew install colmap
```

If using Windows, we recommend you use the native Windows binary rather than the linux one through WSL2 to remove emulation overhead.

Now that COLMAP has been installed, we can start interacting with COLMAP through its GUI[1]. To get COLMAP to run in GUI mode, type:

[1] Note that I think the GUI is pretty cumbersome and buggy. I got it to crash at least five times while writing this tutorial. COLMAP on the command line, by contrast, seems to be as tough as nails. [SJJ]

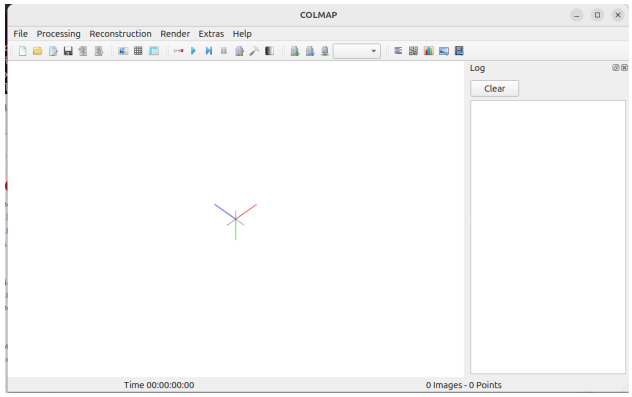

Figure 1: The COLMAP GUI.

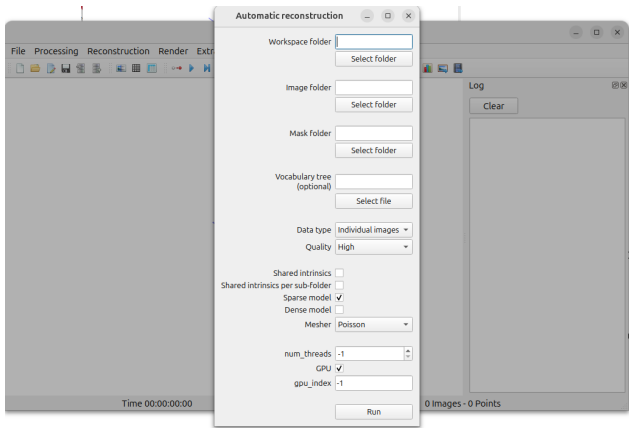

Figure 2: Popup window with options for autoreconstruct.

```
colmap gui
```

should see a window open similar to that shown in Figure 1.

Activity 1: Build a Model of the South Building

In this activity, we'll download a standard well-behaved COLMAP dataset and build the 3D model. Download and unzip the `South Building` **dataset from:**

```
https://demuc.de/colmap/datasets/
```

This will create the directory `south-building`**. Within it you will find** `database.db` **(a cached set of extracted and matched SIFT features), together with the subdirectory** `images` **which contains the raw images. Note you'll need about 800MB of disk space — the zip file is about 400MB, and you'll need another 400MB for the uncompressed contents.**

We'll run COLMAP's "auto reconstruct" pipeline. You can access this either from the icon which looks like a white circle pointing to a red one, or from the "Reconstruction" dropdown menu.

Once selected, you should be presented with a pop up window similar to what you see in Figure 2. The two main things you need to set are the "Workspace" folder to `south-building`**, the image directory**

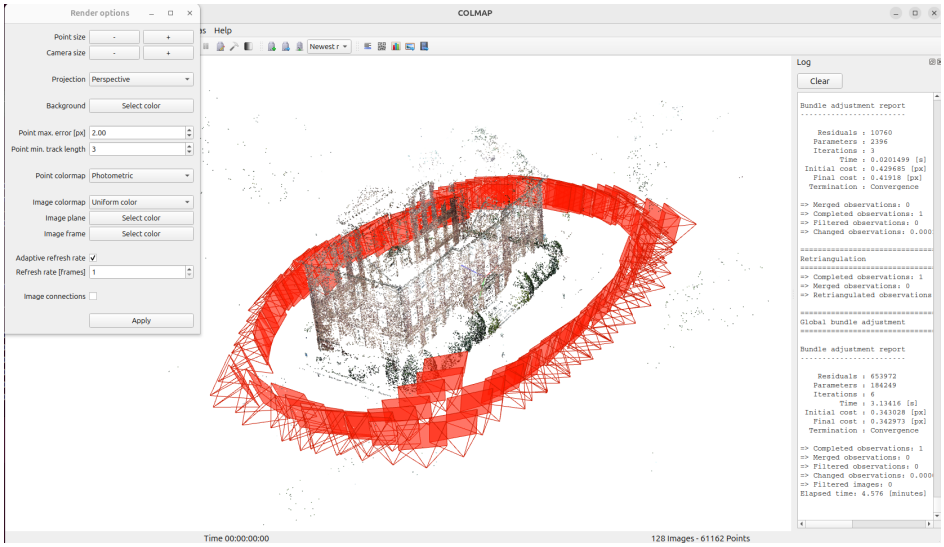

Figure 3: The 3D reconstruction from the `southern-building` dataset. Also shown is a GUI for modifying the display.

to `south-building/images` and the "Image Quality" to medium. This menu lets you set lots of other options (e.g, the vocabularly file for loop closing) but we don't need to touch any values at the moment. Wheu you click "Run", the sparse reconstruction pipeline will run from start to finish. If the database file is used, COLMAP will start with the reconstruction / bundle adjustment. If the database file is not present, it starts with low-level feature extraction and matching. The results should (hopefully) look like the output in Figure 3.

Activity 2: Analyse What COLMAP Did

The automatic reconstruction pipeline is pretty blackbox, but we can interrogate the system to figure out what happened.

COLMAP generates detailed text-based information on its progression. This is shown in the "Log" section in Figure 3 and is also printed on the console (if launched from the command line). If you scroll through this, you should be able to see different activities starting at different times.

By checking the log / terminal output, can you see:

1. What is the default camera model COLMAP uses?
2. How many input features were read and how long did it take to extract SIFT features from all of them?
3.

COLMAP also has a way to check stuff using the GUI, but it's a bit of a pain. You have to start by creating a project (automatic reconstruction doesn't do this). To do this:

- Open the "File" menu and click "New Project". This will open a popup menu asking you to specify the database and the images.
- For the database, select "Open" and then use the file browser to select `database.db`.

- For the images select the `images` subdirectory.
- Press "Save" to save

You can now go to the dropdown menu "Processing" and select "Database Management". This will open a window. You can explore the images and their matches. You have to select an image. If you use the button "Show Image", it will show you the image and the detected features. If you select "Overlapping image", it will show you all the image pairs which have been matched as overlap candidates[2].

COLMAP documents all of its file formats. These can be found at:

```
https://colmap.github.io/database.html
```

Writing a parser for this data is actually very easy (especially with your code generator of choice). As a bonus, you could look at reading these files to see if you can extract the scenegraph data (pairwise match types and strengths) and information such as a histogram of track lengths.

Activity 3: Go Forth and Do Some SfM-ing

In this activity, the goal is to start collecting your own datasets to see if you can get COLMAP to reconstruct them. This is deliberately experimental and vague; in my experience the best way is to play with stuff and find what works and what doesn't work.

For this lab, we suggest you do not use video but, rather, go out and take photos using a camera. The reason why is you have to think about where to stand next and where to point the camera — when walking it's much more constrained.

You can follow a similar pattern to the `southern-building` dataset: create a top-level directory, a subdirectory called `images` and copy the images there.

Some suggested activities:

1. Attempt to model something small, such as a computer or a desk, in the innovation lab. This will require a small number of frames and will be processed quickly.
2. Be more ambitious and try to map more of the Innovation Lab. For example, can you model the desks and the space between them?
3. Try to model the space outside the innovation lab with the tables. How hard is it? Why do you think the challenges might be different from operating inside the Innovation Lab?
4. Try varying the camera settings such as the zoom while modelling a single object. How successful is COLMAP at handling changes to camera intrinsics?
5. Use several different cameras on, say, several different phones and try to combine all the images together.
6. Try modelling a much larger structure. Can you, for example, model the exterior of Marshgate (although be careful with camera theft, etc.)
7. Try to make a dataset which works "sparser" by removing frames from the `images` subdirectory and running COLMAP from scratch again.

We're very happy to talk about your results and your experiences with the software and to discuss further ideas.

[2] However, for me various fields, such as `num_matches` are always blank [SJJ].